

01:XXX:XXX - Homework n

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Problem 1

Prove:

1. $\mathbb{R} \times \mathbb{R} \sim \mathbb{R}$.
2. $\mathbb{R} \setminus \mathbb{N} \sim \mathbb{R}$.
3. $\mathbb{R} \setminus \mathbb{Q} \sim \mathbb{R}$.

Proof. We use that $\mathbb{R} \sim \{0, 1\}^{\mathbb{N}}$.

(1) Since $\mathbb{R} \sim \{0, 1\}^{\mathbb{N}}$, we have

$$\mathbb{R} \times \mathbb{R} \sim \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}}.$$

Define a map

$$\Phi : \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}} \rightarrow \{0, 1\}^{\mathbb{N}}$$

by interleaving: for sequences $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$, set

$$c_{2n} = a_n, \quad c_{2n+1} = b_n \quad \text{for all } n \in \mathbb{N},$$

and define $\Phi((a_n), (b_n)) = (c_k)_{k \in \mathbb{N}}$.

This map is a bijection: given (c_k) , we recover (a_n) and (b_n) by $a_n = c_{2n}$ and $b_n = c_{2n+1}$. Hence

$$\{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}} \sim \{0, 1\}^{\mathbb{N}} \sim \mathbb{R},$$

so $\mathbb{R} \times \mathbb{R} \sim \mathbb{R}$.

(2) Since $\mathbb{N}$ is countable and $\mathbb{R}$ is uncountable, we clearly have

$$\mathbb{R} \setminus \mathbb{N} \subseteq \mathbb{R}$$

via the inclusion map. For the other direction, enumerate $\mathbb{N}$ as $\{n_0, n_1, n_2, \dots\}$ and define a map $f : \mathbb{R} \rightarrow \mathbb{R} \setminus \mathbb{N}$ by

$$f(x) = \begin{cases} x, & \text{if } x \notin \mathbb{N}, \\ k, & \text{if } x = n_k \text{ (so } k \in \mathbb{N}). \end{cases}$$

This is well-defined and injective: distinct reals either coincide as non-naturals where f is the identity, or they are distinct naturals $n_j \neq n_k$ and then $f(n_j) = j \neq k = f(n_k)$.

Thus we have injections

$$\mathbb{R} \setminus \mathbb{N} \subseteq \mathbb{R} \quad \text{and} \quad \mathbb{R} \subseteq \mathbb{R} \setminus \mathbb{N},$$

so by the Cantor–Bernstein–Schröder theorem,

$$\mathbb{R} \setminus \mathbb{N} \sim \mathbb{R}.$$

(3) The set $\mathbb{Q}$ is countable, so the same argument applies with $\mathbb{Q}$ in place of $\mathbb{N}$. Therefore removing a countable subset from $\mathbb{R}$ does not change its cardinality, and

$$\mathbb{R} \setminus \mathbb{Q} \sim \mathbb{R}.$$

□

Problem 2

Let A be the set of all bijections $f : \mathbb{N} \rightarrow \mathbb{N}$. Show that $A \sim \{0, 1\}^{\mathbb{N}}$.

Proof. We construct injections both ways and apply the Cantor–Bernstein–Schröder theorem.

Injection $\{0, 1\}^{\mathbb{N}} \rightarrow A$. Given $x = (x_n)_{n \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}}$, define $f_x : \mathbb{N} \rightarrow \mathbb{N}$ as follows. For each $n \in \mathbb{N}$, consider the pair $\{2n, 2n + 1\}$:

- If $x_n = 0$, let $f_x(2n) = 2n$ and $f_x(2n + 1) = 2n + 1$.
- If $x_n = 1$, let $f_x(2n) = 2n + 1$ and $f_x(2n + 1) = 2n$.

This defines a permutation of $\mathbb{N}$, hence a bijection, so $f_x \in A$. If $x \neq y$, there is some n with $x_n \neq y_n$, and on the pair $\{2n, 2n + 1\}$ the maps f_x and f_y act differently. Thus $f_x \neq f_y$. Therefore

$$\Phi : \{0, 1\}^{\mathbb{N}} \rightarrow A, \quad \Phi(x) = f_x$$

is injective.

Injection $A \rightarrow \{0, 1\}^{\mathbb{N}}$. We know from class that $\mathbb{N}^{\mathbb{N}} \sim \{0, 1\}^{\mathbb{N}}$. Since $A \subseteq \mathbb{N}^{\mathbb{N}}$, the inclusion map

$$\iota : A \hookrightarrow \mathbb{N}^{\mathbb{N}}$$

is injective. Composing ι with a fixed bijection $\psi : \mathbb{N}^{\mathbb{N}} \rightarrow \{0, 1\}^{\mathbb{N}}$ gives an injection

$$\psi \circ \iota : A \rightarrow \{0, 1\}^{\mathbb{N}}.$$

We now have

$$\{0, 1\}^{\mathbb{N}} \preceq A \quad \text{and} \quad A \preceq \{0, 1\}^{\mathbb{N}},$$

so by Cantor–Bernstein–Schröder,

$$A \sim \{0, 1\}^{\mathbb{N}}.$$

□

Problem 3

For sets A, B let $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

- Show $A = B$ iff $A \Delta B = \emptyset$.
- Show $(A \Delta B) \cup (B \Delta C) \subseteq A \Delta C$.
- Let

$$R = \{\langle A, B \rangle \in \mathcal{P}(\mathbb{N})^2 : 2026 \notin A \Delta B\}.$$

Show R is an equivalence relation on $\mathcal{P}(\mathbb{N})$.

- Compute $[\emptyset]_R$, $[\mathbb{N}]_R$ and the number of equivalence classes.

Proof. (i) Suppose $A = B$. Then $A \setminus B = \emptyset$ and $B \setminus A = \emptyset$, so $A \Delta B = \emptyset \cup \emptyset = \emptyset$.

Conversely, if $A \Delta B = \emptyset$, then no element lies in exactly one of A, B . Hence every element of A lies in B and every element of B lies in A , so $A = B$.

(ii) For any set X and element x , let $1_X(x)$ be 1 if $x \in X$ and 0 otherwise. Then $x \in A \Delta B$ iff $1_A(x) \oplus 1_B(x) = 1$, where $\oplus$ is addition modulo 2. Similarly, $x \in B \Delta C$ iff $1_B(x) \oplus 1_C(x) = 1$, and $x \in A \Delta C$ iff $1_A(x) \oplus 1_C(x) = 1$.

Suppose $x \in (A \Delta B) \cup (B \Delta C)$. Then at least one of $1_A(x) \oplus 1_B(x)$ and $1_B(x) \oplus 1_C(x)$ equals 1. But in $\{0, 1\}$ we have

$$(1_A(x) \oplus 1_B(x)) \oplus (1_B(x) \oplus 1_C(x)) = 1_A(x) \oplus 1_C(x).$$

If at least one of the two terms on the left is 1, then the right-hand side is 1, so $1_A(x) \oplus 1_C(x) = 1$. Hence $x \in A \Delta C$. Therefore $(A \Delta B) \cup (B \Delta C) \subseteq A \Delta C$.

(iii) We verify that R is an equivalence relation on $\mathcal{P}(\mathbb{N})$.

Reflexive: For every $A \subseteq \mathbb{N}$ we have $A \Delta A = \emptyset$, so in particular $2026 \notin A \Delta A$. Hence $\langle A, A \rangle \in R$.

Symmetric: If $\langle A, B \rangle \in R$, then $2026 \notin A \Delta B$. But $A \Delta B = B \Delta A$, so $2026 \notin B \Delta A$, and thus $\langle B, A \rangle \in R$.

Transitive: Suppose $\langle A, B \rangle \in R$ and $\langle B, C \rangle \in R$. Then $2026 \notin A \Delta B$ and $2026 \notin B \Delta C$, which means 2026 is in both A and B or in neither, and also in both B and C or in neither. Hence 2026 is in A and C together or in neither, so $2026 \notin A \Delta C$, and $\langle A, C \rangle \in R$.

Thus R is an equivalence relation.

(iv) By definition, A and B are R -equivalent iff 2026 is either in both A, B or in neither. Therefore

$$[\emptyset]_R = \{A \subseteq \mathbb{N} : 2026 \notin A\},$$

and

$$[\mathbb{N}]_R = \{A \subseteq \mathbb{N} : 2026 \in A\}.$$

There are exactly two equivalence classes: all subsets of $\mathbb{N}$ that do not contain 2026, and all subsets that do contain 2026. $\square$

Problem 4 (Optional)

On $\{0, 1\}^{\mathbb{N}}$, define $f \sim g$ iff $f(0) = g(0)$, $f(1) = g(1)$, and $f(2) = g(2)$. How many different equivalence classes are there?

Proof. Two functions $f, g \in \{0, 1\}^{\mathbb{N}}$ are equivalent iff they agree on the first three coordinates. Thus each equivalence class is determined by the triple

$$(f(0), f(1), f(2)) \in \{0, 1\}^3.$$

There are $2^3 = 8$ possible such triples, hence there are 8 equivalence classes. $\square$

Problem 5

Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections, and let $h = g \circ f : A \rightarrow A$. We construct a bijection $k : A \rightarrow \text{rng}(g)$ and then $g^{-1} \circ k : A \rightarrow B$, obtaining the Cantor–Bernstein–Schröder theorem. As an application, we get a bijection between $[0, 1)$ and $[0, 1]$.

Proof. (I) Since f and g are injections, their composition $h = g \circ f$ is also injective.

(II) The map $g : B \rightarrow \text{rng}(g)$ is a bijection by definition of the range, hence it has an inverse $g^{-1} : \text{rng}(g) \rightarrow B$. Thus, if we construct a bijection $k : A \rightarrow \text{rng}(g)$, then

$$g^{-1} \circ k : A \rightarrow B$$

is a bijection as the composition of bijections.

(III) Define subsets $A_0, A_1, \dots$ of A recursively by

$$A_0 = A \setminus \text{rng}(g), \quad A_{n+1} = h[A_n] \quad \text{for all } n \in \mathbb{N}.$$

Since h is injective, the restriction $h \upharpoonright A_n : A_n \rightarrow A_{n+1}$ is a bijection from A_n onto A_{n+1} for each n . The sets A_n are pairwise disjoint by construction.

(IV) Define $k : A \rightarrow \text{rng}(g)$ by

$$k(a) = \begin{cases} h(a), & \text{if } a \in A_n \text{ for some } n, \\ a, & \text{if } a \notin \bigcup_{n \in \mathbb{N}} A_n. \end{cases}$$

If $a \in A_n$, then $h(a) \in A_{n+1} \subseteq A \subseteq \text{rng}(g) \cup A_0$, and by construction $A_0 \subseteq A \setminus \text{rng}(g)$, so in fact $h(a) \in \text{rng}(g)$. If $a \notin \bigcup_n A_n$, then by definition $a \in \text{rng}(g)$. Thus $k(a) \in \text{rng}(g)$ in either case, so k is well-defined with codomain $\text{rng}(g)$.

(V) To see that k is injective, note first that

for any $a \in A$, $k(a) \in A_{n+1}$ for some n iff $a \in A_n$ for some n .

This uses that $h[A_n] = A_{n+1}$ and that the A_n are pairwise disjoint. Now suppose $k(a) = k(a') = b$. If $b \in A_{m+1}$ for some m , then by the claim both a, a' lie in A_m , and on A_m the map k coincides with h , which is injective, so $a = a'$. If $b \notin A_n$ for any n , then b is fixed by k , so $a = b = a'$. Thus k is injective.

(VI) To see that k is surjective onto $\text{rng}(g)$, let $y \in \text{rng}(g)$. If $y \notin A_{n+1}$ for any n , then by definition $k(y) = y$, so y is in the range of k . If $y \in A_{n+1}$ for some n , then there is a unique $x \in A_n$ with $h(x) = y$, because $h \upharpoonright A_n$ is a bijection onto A_{n+1} . For this x we have $k(x) = h(x) = y$, so again y is in the range of k . Therefore $k : A \rightarrow \text{rng}(g)$ is a bijection, and thus $g^{-1} \circ k : A \rightarrow B$ is a bijection.

This proves the Cantor–Bernstein–Schröder theorem.

Application: a bijection $[0, 1) \rightarrow [0, 1]$.

Define injections

$$f_0 : [0, 1) \rightarrow [0, 1], \quad f_0(x) = x,$$

and

$$g_0 : [0, 1] \rightarrow [0, 1), \quad g_0(x) = \frac{x}{2}.$$

Both f_0 and g_0 are injective. By the Cantor–Bernstein–Schröder theorem, there exists a bijection between $[0, 1)$ and $[0, 1]$. $\square$